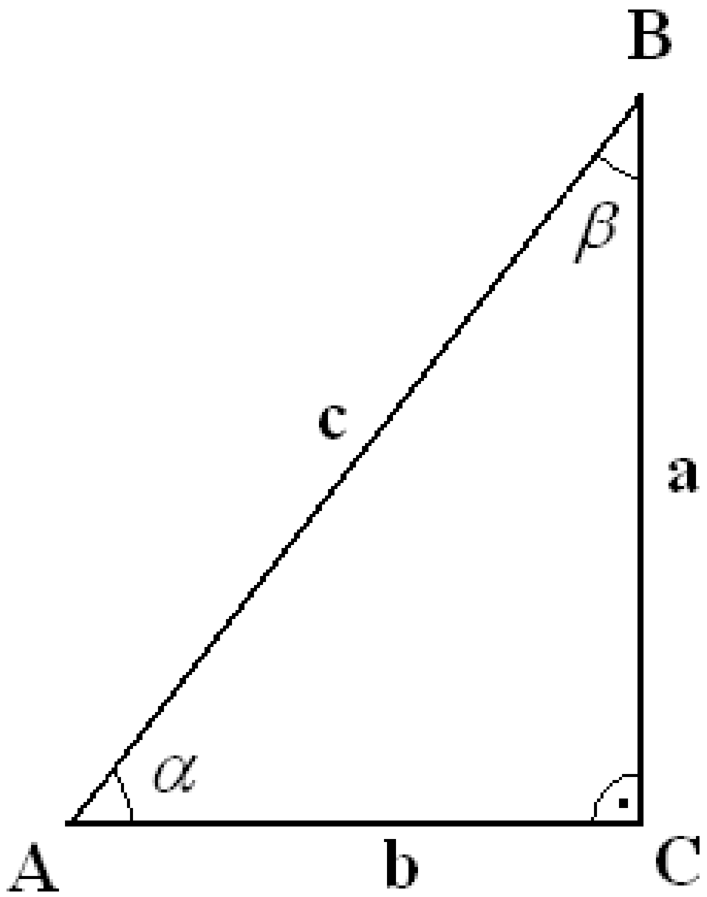


$$\begin{cases} \sin\alpha = \dfrac{a}{c} \\ \cos\alpha = \dfrac{b}{c} \\ tg\alpha = \dfrac{a}{b} \\ \cot g\alpha = \dfrac{b}{a} \end{cases} \qquad \begin{cases} \sin\beta = \dfrac{b}{c} \\ \cos\beta = \dfrac{a}{c} \\ tg\beta = \dfrac{b}{a} \\ \cot g\beta = \dfrac{a}{b} \end{cases}$$

Квадрант:	**sin α**	**cos α**	**tg α**	**cotg α**
I	+	+	+	+
II	+	-	-	-
III	-	-	+	+
IV	-	+	-	-

α	0°	30°	45°	60°	90°	120°	150°	180°	270°	360°
sin α	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	0	-1	0
cos α	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	0	1
tg α	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$		$-\sqrt{3}$	$-\frac{\sqrt{3}}{3}$	0		0
cotg α		$\sqrt{3}$	1	$\frac{\sqrt{3}}{3}$	0	$-\frac{\sqrt{3}}{3}$	$-\sqrt{3}$		0	

Основни тъждества

$\sin^2 \alpha + \cos^2 \alpha = 1$

$\mathrm{tg}\,\alpha = \dfrac{\sin\alpha}{\cos\alpha}$

$1 + \mathrm{tg}^2\,\alpha = \dfrac{1}{\cos^2\alpha}$

$\mathrm{tg}\,\alpha \,.\, \mathrm{cotg}\,\alpha = 1$

$\mathrm{cotg}\,\alpha = \dfrac{\cos\alpha}{\sin\alpha}$

$1 + \mathrm{cotg}^2\,\alpha = \dfrac{1}{\sin^2\alpha}$

Връзка между тригонометричните функции

	sin α	**cos α**	**tg α**	**cotg α**
sin α =	sin α	$\pm\sqrt{1-\cos^2\alpha}$	$\dfrac{tg\alpha}{\pm\sqrt{1+tg^2\alpha}}$	$\dfrac{1}{\pm\sqrt{1+\cot g^2\alpha}}$
cos α =	$\pm\sqrt{1-\sin^2\alpha}$	cos α	$\dfrac{1}{\pm\sqrt{1+tg^2\alpha}}$	$\dfrac{\cot g\alpha}{\pm\sqrt{1+\cot g^2\alpha}}$
tg α =	$\dfrac{\sin\alpha}{\pm\sqrt{1-\sin^2\alpha}}$	$\dfrac{\pm\sqrt{1-\cos^2\alpha}}{\cos^2\alpha}$	tg α	$\dfrac{1}{\cot g\alpha}$
cotg α =	$\dfrac{\pm\sqrt{1-\sin^2\alpha}}{\sin^2\alpha}$	$\dfrac{\cos\alpha}{\pm\sqrt{1-\cos^2\alpha}}$	$\dfrac{1}{tg\alpha}$	cotg α

	$-\alpha$	$90° - \alpha$	$90° + \alpha$	$180° - \alpha$	$180° + \alpha$	$270° - \alpha$	$270° + \alpha$	$k.360° - \alpha$	$k.360° + \alpha$
sin α	- sin α	+ cos α	+ cos α	+ sin α	- sin α	- cos α	- cos α	- sin α	+ sin α
cos α	+ cos α	+ sin α	- sin α	- cos α	- cos α	- sin α	+ sin α	+ cos α	+ cos α
tg α	- tg α	+cotgα	-cotgα	- tg α	+ tg α	+cotgα	-cotgα	- tg α	+ tg α
cotg α	-cotgα	+ tg α	- tg α	-cotgα	+cotgα	+ tg α	- tg α	-cotgα	+cotgα

Синусова теорема

Сбор и разлика $\frac{a}{\sin\alpha}=\frac{b}{\sin\beta}=\frac{c}{\sin\gamma}=2R$ на 2 ъгъла

$\sin(\alpha\pm\beta)=\sin\alpha.\cos\beta\pm\cos\alpha.\sin\beta$

$\cos(\alpha\pm\beta)=\cos\alpha.\cos\beta\mp\sin\alpha.\sin\beta$

$tg(\alpha\pm\beta)=\frac{tg\alpha\pm tg\beta}{1\mp tg\alpha.tg\beta}$

$cotg(\alpha\pm\beta)=\frac{\cot g\alpha.\cot g\beta\mp 1}{\cot g\beta\pm\cot g\alpha}$

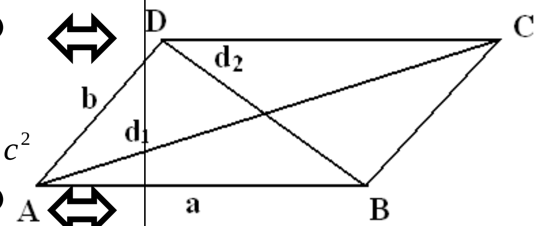
Косинусова теорема
$a^2=b^2+c^2-2bc.\cos\alpha$ $b^2=a^2+c^2-2ac.\cos\beta$ $c^2=a^2+b^2-2ab.\cos\gamma$

Следствие 1
$a=\frac{c.\sin\alpha}{\sin(\alpha+\beta)}$ $b=\frac{c.\sin\beta}{\sin(\alpha+\beta)}$ $c=\frac{a.\sin(\alpha+\beta)}{\sin\alpha}$

Следствие 2
$\sin\alpha=\frac{a}{b}\sin\beta$ $\sin\beta=\frac{b}{a}\sin\alpha$ $\sin\gamma=\frac{c}{b}\sin\beta$

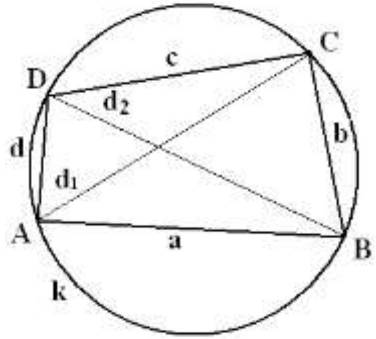
Следствие 3
$\cos\alpha=\frac{b^2+c^2-a^2}{2bc}$ $\cos\beta=\frac{a^2+c^2-b^2}{2ac}$ $\cos\gamma=\frac{a^2+b^2-c^2}{2ab}$

Следствие 4
$\gamma<90^\circ \Leftrightarrow a^2+b^2>c^2$ $\gamma=90^\circ \Leftrightarrow a^2+b^2=c^2$ $\gamma>90^\circ \Leftrightarrow a^2+b^2<c^2$



В успоредник $2(a^2+b^2)=d_1^2+d_2^2$

Произволен четириъгълник, вписан в k(O,R)

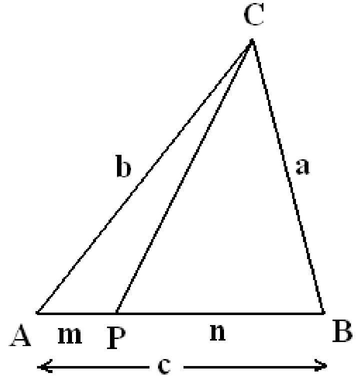


Условие: $AB+CD=AD+BC$

I Теорема на Птолемей: $d_1d_2=ac+bd$

II Теорема на Птолемей: $\frac{d_1}{d_2}=\frac{ab+cd}{bc+ad}$

Теорема на Стюарт



$$CP^2 = \frac{ma^2 + nb^2}{c} - mn$$

Ако P е продължение на страната AB =>

$$\begin{cases} CP^2 = \dfrac{-ma^2 + nb^2}{c} + mn \\ CP^2 = \dfrac{ma^2 - nb^2}{c} + mn \end{cases}$$

, когато A е между P и B;

, когато B е между A и P.

Формули за медианите в триъгълника

$m_a = \frac{1}{2}\sqrt{2(b^2 + c^2) - a^2}$ $m_b = \frac{1}{2}\sqrt{2(a^2 + c^2) - b^2}$ $m_c = \frac{1}{2}\sqrt{2(a^2 + b^2) - c^2}$

Формули за ъглополовящите в триъгълника

$$p=\frac{a+b+c}{2} \quad => \quad l_a=\frac{2}{b+c}\sqrt{bcp(p-a)} \qquad l_b=\frac{2}{a+c}\sqrt{acp(p-b)} \qquad l_c=\frac{2}{a+b}\sqrt{abp(p-c)}$$

Формула на Ойлер $d=\sqrt{R(R-2r)}$

Ако M е медицентър => $a^2 + b^2 + c^2 = 3(MA^2 + MB^2 + MC^2)$

Теорема на Лайбниц $PA^2 + PB^2 + PC^2 = MA^2 + MB^2 + MC^2 + 3.PM^2$

където P е произволна точка (вътре, или извън триъгълника ABC).

Тангенсова Теорема $\frac{a+b}{a-b}=\frac{tg\frac{a+b}{2}}{tg\frac{a-b}{2}}$

Двойни и половинки ъгли

$$\sin 2\alpha = 2\sin\alpha.\cos\alpha$$

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha = \frac{1}{2} - 2\sin^2\alpha = 2\cos^2\alpha - 1$$

$$\cos\alpha \pm \sin\alpha = \sqrt{1 \pm \sin 2\alpha}$$

$$\sin\frac{\alpha}{2} = \pm\sqrt{\frac{1-\cos\alpha}{2}} \qquad \cos\frac{\alpha}{2} = \pm\sqrt{\frac{1+\cos\alpha}{2}} \qquad \sin\alpha = \frac{2tg\frac{\alpha}{2}}{1+tg^2\frac{\alpha}{2}} \qquad \cos\alpha = \frac{1-tg^2\frac{\alpha}{2}}{1+tg^2\frac{\alpha}{2}}$$

Понижаване на степен

$$2\sin^2\alpha = 1-\cos 2\alpha \qquad 2\cos^2\alpha = 1+\cos 2\alpha$$

Други

$$\sin\alpha + \sin\beta = 2\sin\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2} \qquad \sin\alpha - \sin\beta = 2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} \qquad \cos\alpha + \cos\beta = 2\cos\frac{\alpha+\beta}{2}\cos\frac{\alpha-\beta}{2}$$

$$\cos\alpha - \cos\beta = -2\sin\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2} = 2\sin\frac{\alpha+\beta}{2}\sin\frac{\beta-\alpha}{2}$$